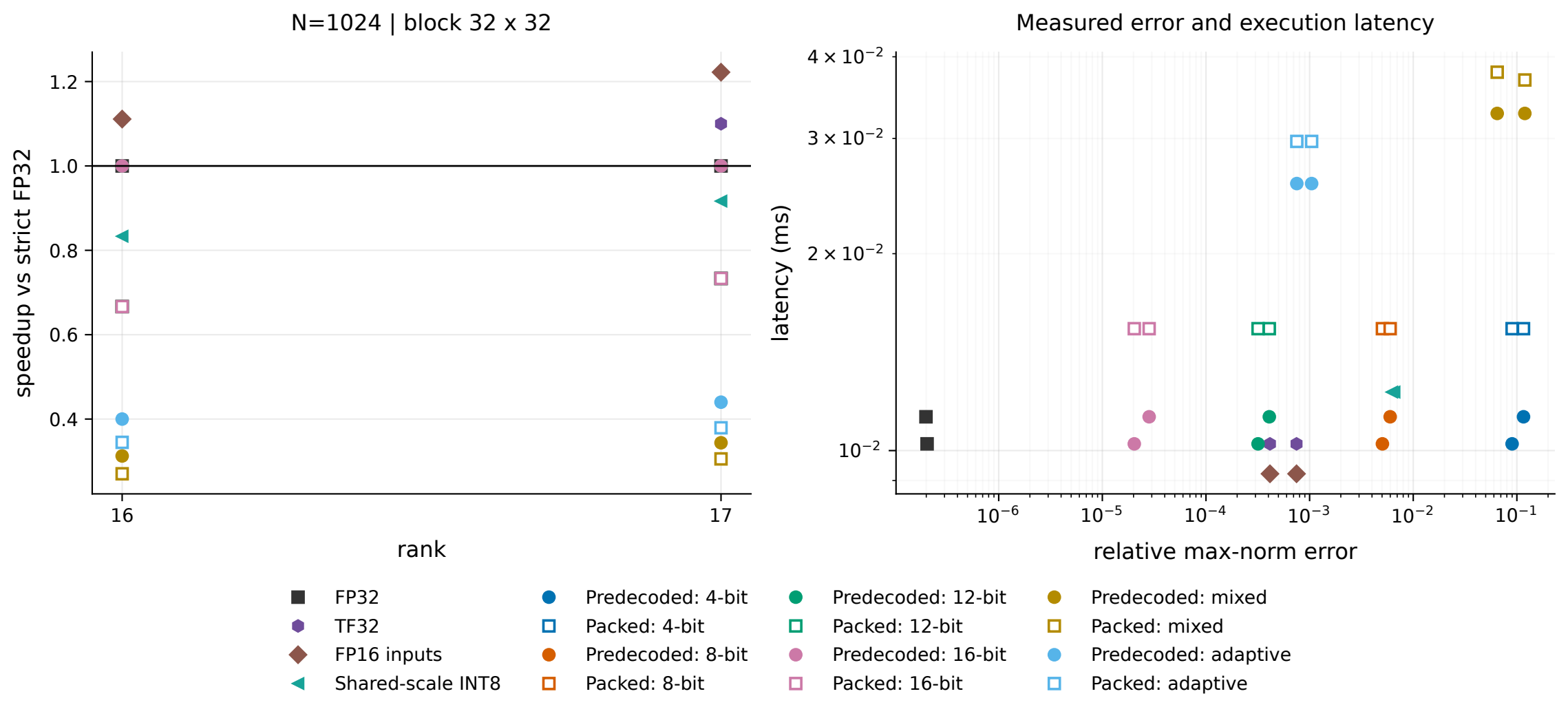
N=1024 | block 32 x 32
speedup vs strict FP32
1.2
1.0
0.8
0.6
0.4
16
17
rank
Measured error and execution latency
latency (ms)
4 × 10⁻²
3 × 10⁻²
2 × 10⁻²
10⁻²
10⁻⁶
10⁻⁵
10⁻⁴
10⁻³
10⁻²
10⁻¹
relative max-norm error
FP32
TF32
FP16 inputs
Shared-scale INT8
Predecoded: 4-bit
Packed: 4-bit
Predecoded: 8-bit
Packed: 8-bit
Predecoded: 12-bit
Packed: 12-bit
Predecoded: 16-bit
Packed: 16-bit
Predecoded: mixed
Packed: mixed
Predecoded: adaptive
Packed: adaptive